

NETLIST $\rightarrow$ List of Networks $\rightarrow$

Network

* In a Simulator;

- LTspice
- Xschem
- Cadence
- Synopsys

NOTE ①
At different Nodes;
we have different
Potential differences.

are
always
RELATIVE !!
to components

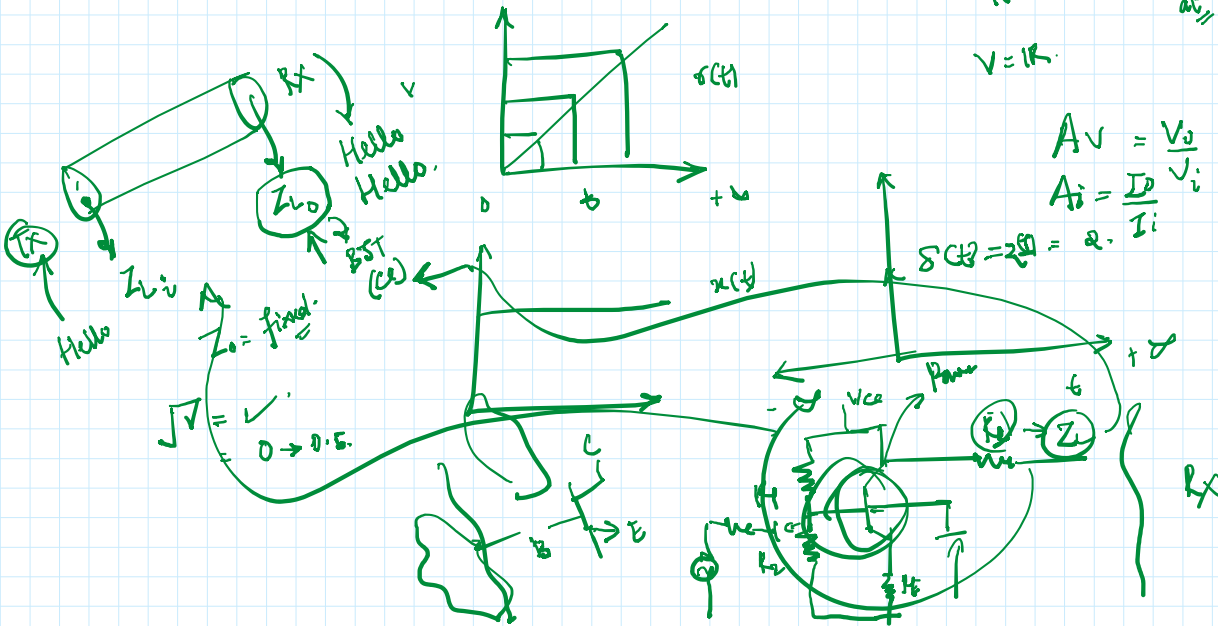
② "NETLISTS" & "MODEL FILES" of various
devices/components/I.Gs have a completely
different meaning.

DEVICES/
COMPONENTS

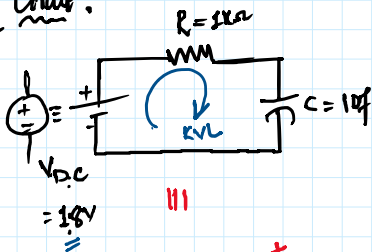
- I. Active device $\rightarrow$ MOSFET, BJT, diode, Op-amp
- II. Passive device $\rightarrow$ R, C, L.
- III. Time dependent $\rightarrow$ C, L
- IV. Time Independent

$V = IR$

$L \frac{di}{dt}, G \frac{dV}{dt}$



1. Circuit $\rightarrow$ $i_c = 0A$



$V_{DC} - V_R - V_C = 0$

$V_{DC} = V_R + V_C$

$V_R = 25V$
 $V_C = 2V$

$V_R = I \cdot R$

$i_c = C \cdot \frac{dV}{dt} = 10f \left(\frac{dV}{dt} \right) = 10f \cdot \frac{d\{const\}}{dt} = 10f \cdot 0 = 0$

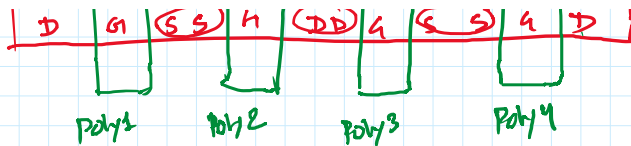
$i_c = 0 \rightarrow R = \infty \rightarrow \text{cut} \rightarrow \text{OPEN}$

$V_C = i_c \cdot 0 = 0$

$V_R = IR$

$V_R = i_c \cdot R = 0 \cdot R = 0V$

$V_R = 0V$

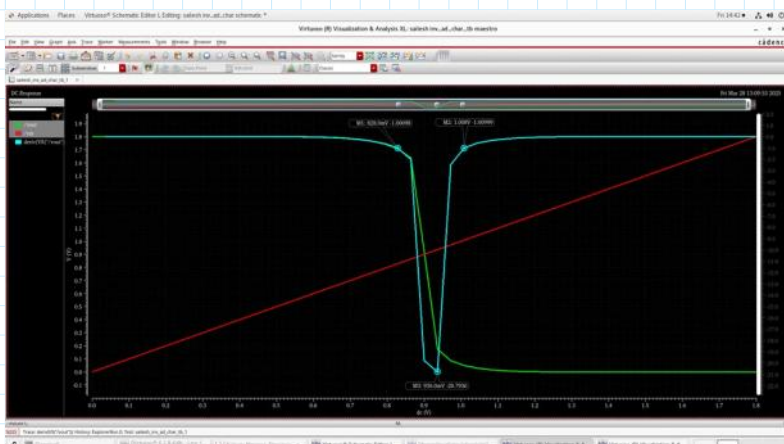
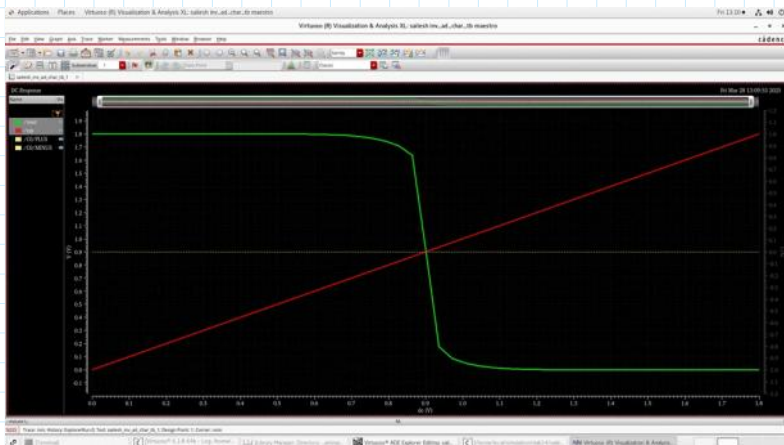
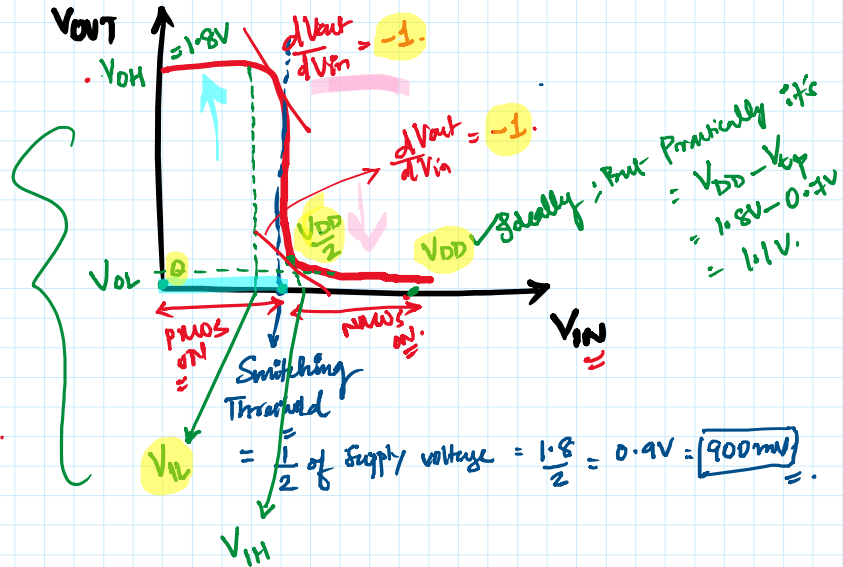


$$\Rightarrow I_D = \frac{1}{2} \mu_n C_{ox} \cdot \frac{W}{L} \left\{ (V_{GS} - V_{th})^2 - \frac{V_{DS}^2}{2} \right\}$$

$$\Rightarrow I_D \propto \left(\frac{W}{L} \right)^2 \rightarrow \text{PMOS}$$

① High: NOISE MARGIN = $V_{OH} - V_{IH}$

② Low: NOISE MARGIN = $V_{IL} - V_{OL}$
 $V_{OL} \text{ (ideally)} = 0V$
 $V_{DD} - V_{th} \text{ (Practically)}$



$$NMH = V_{OH} - V_{IH} = 1.8 - 1.008 = 0.792V$$

$$NML = V_{IL} - V_{OL} = 828.0mV - 0 = 0.792V$$